Hospital Management System in C Language

A console-based solution for automating patient records, staff management, and appointment scheduling — built entirely in C.

ACADEMIC PROJECT

C PROGRAMMING

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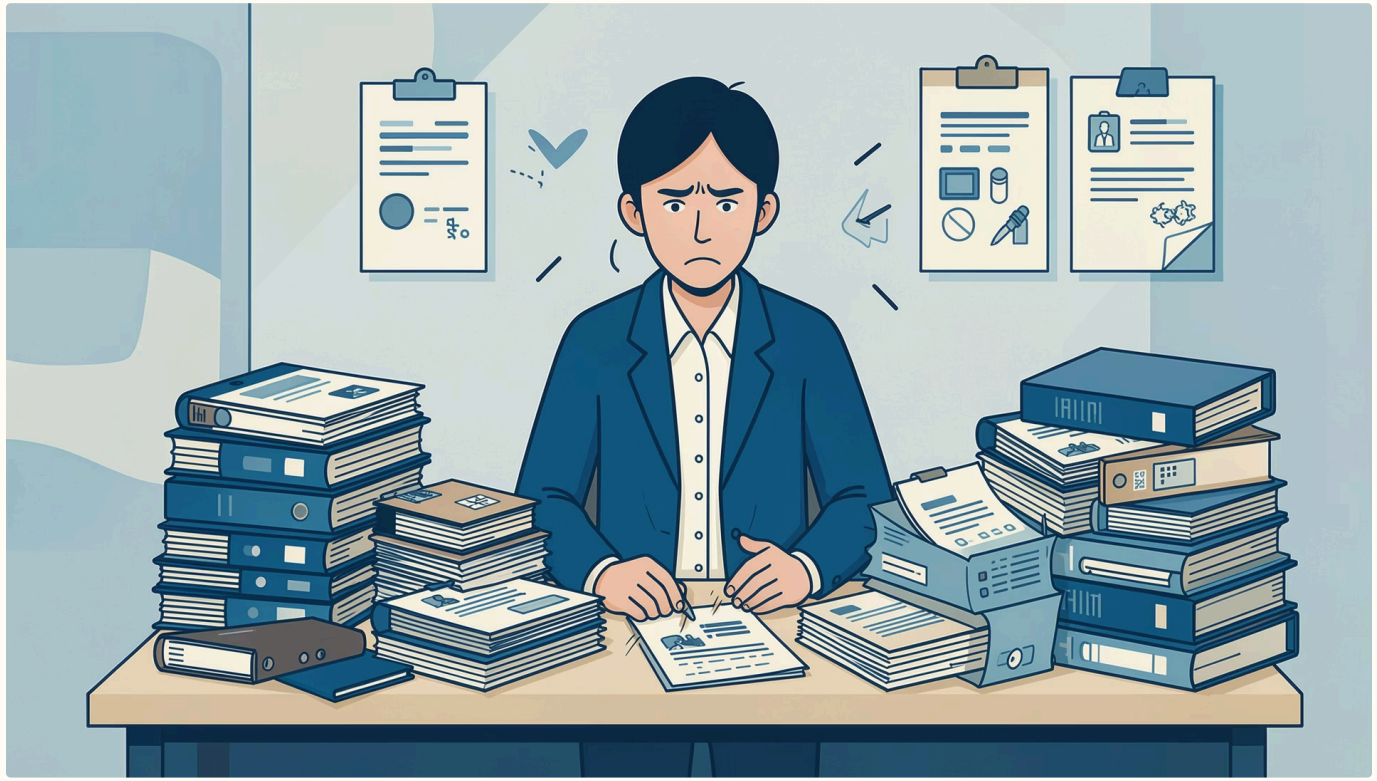
Introduction & Problem Statement



The Challenge

Manual hospital data management is **error-prone, slow, and inefficient**. Paper-based records are vulnerable to loss, duplication, and inconsistency — directly impacting patient care.

Our Solution

A **console-based Hospital Management System in C** that automates patient records, staff management, and appointments — improving accuracy, accessibility, and operational efficiency.

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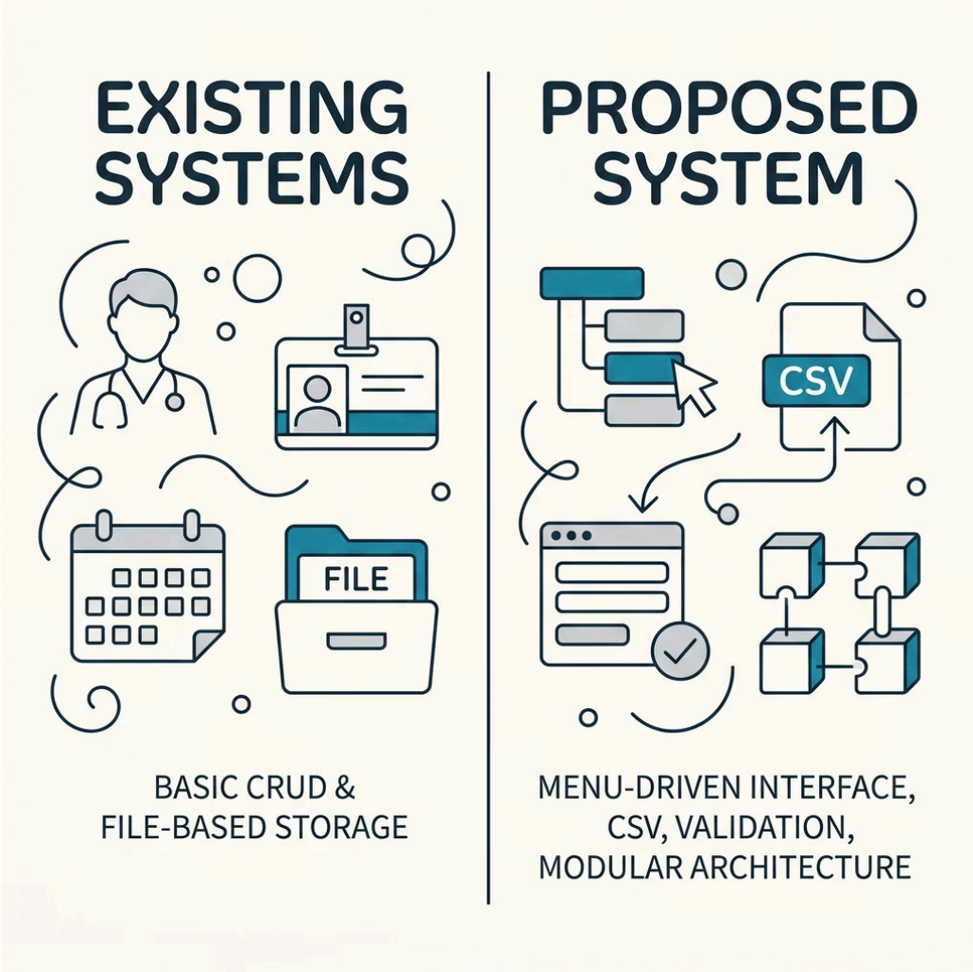
Literature Survey & Proposed System



What We Found

Existing systems rely on CRUD operations for patients, staff, and appointments with basic file storage.

What We Built

Inspired by open-source GitHub projects and GeeksforGeeks tutorials.

Menu-Driven Interface

Intuitive navigation for all user roles

CSV Data Persistence

Records stored reliably across sessions

Input Validation

Robust error handling throughout

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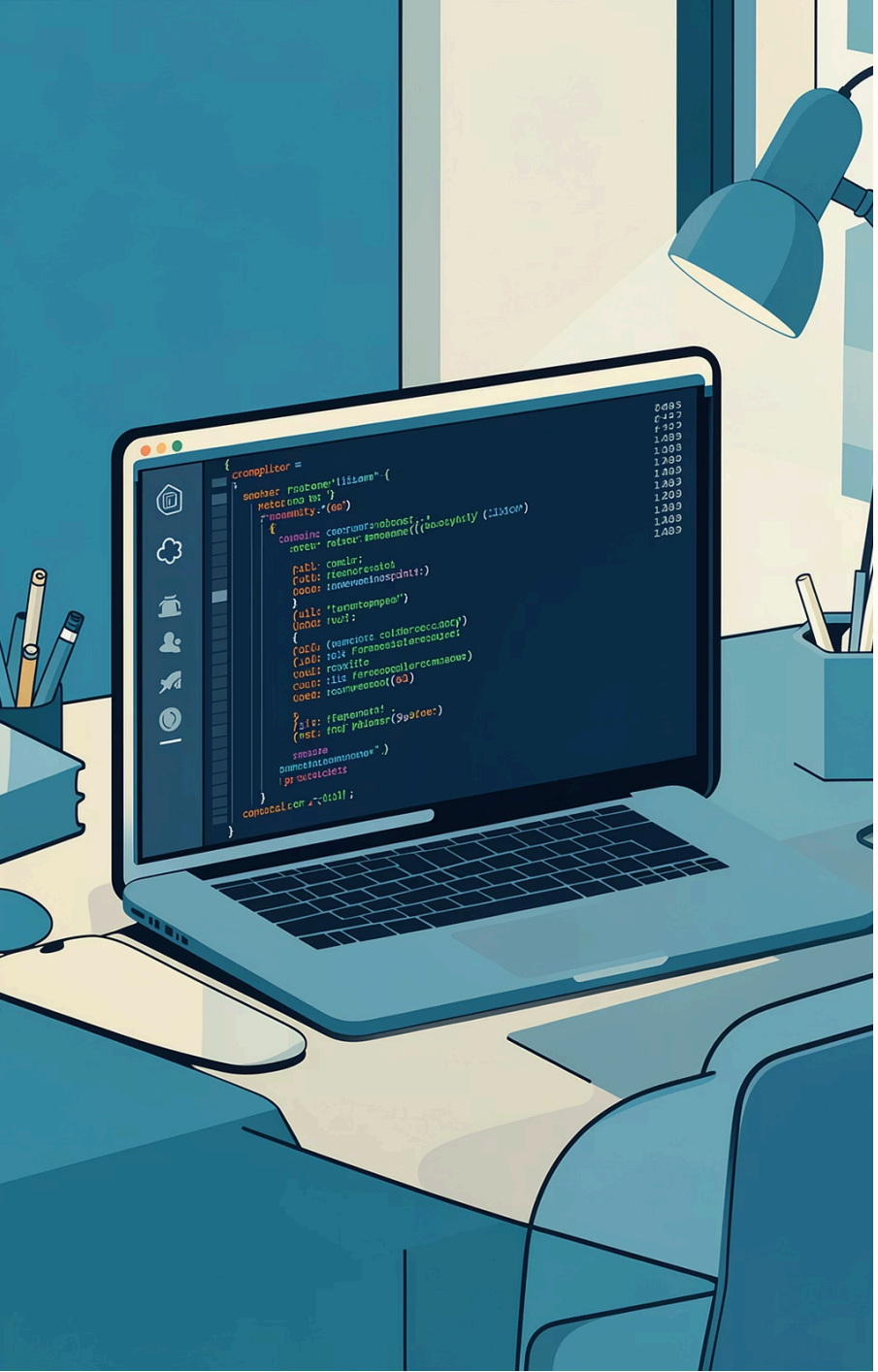

Hardware & Software Requirements

🖥️ Hardware

- Standard PC or Laptop
- Minimum 2GB RAM
- Keyboard & display for console interaction

⚙️ Software

- GCC Compiler (MinGW on Windows)
- Console/Terminal environment
- Optional IDE: Code::Blocks, Dev-C++, VS Code

🔧 Compilation

System compiled with:

```
gcc main.c patient.c staff.c
appointment.c utils.c -o hms.exe
```

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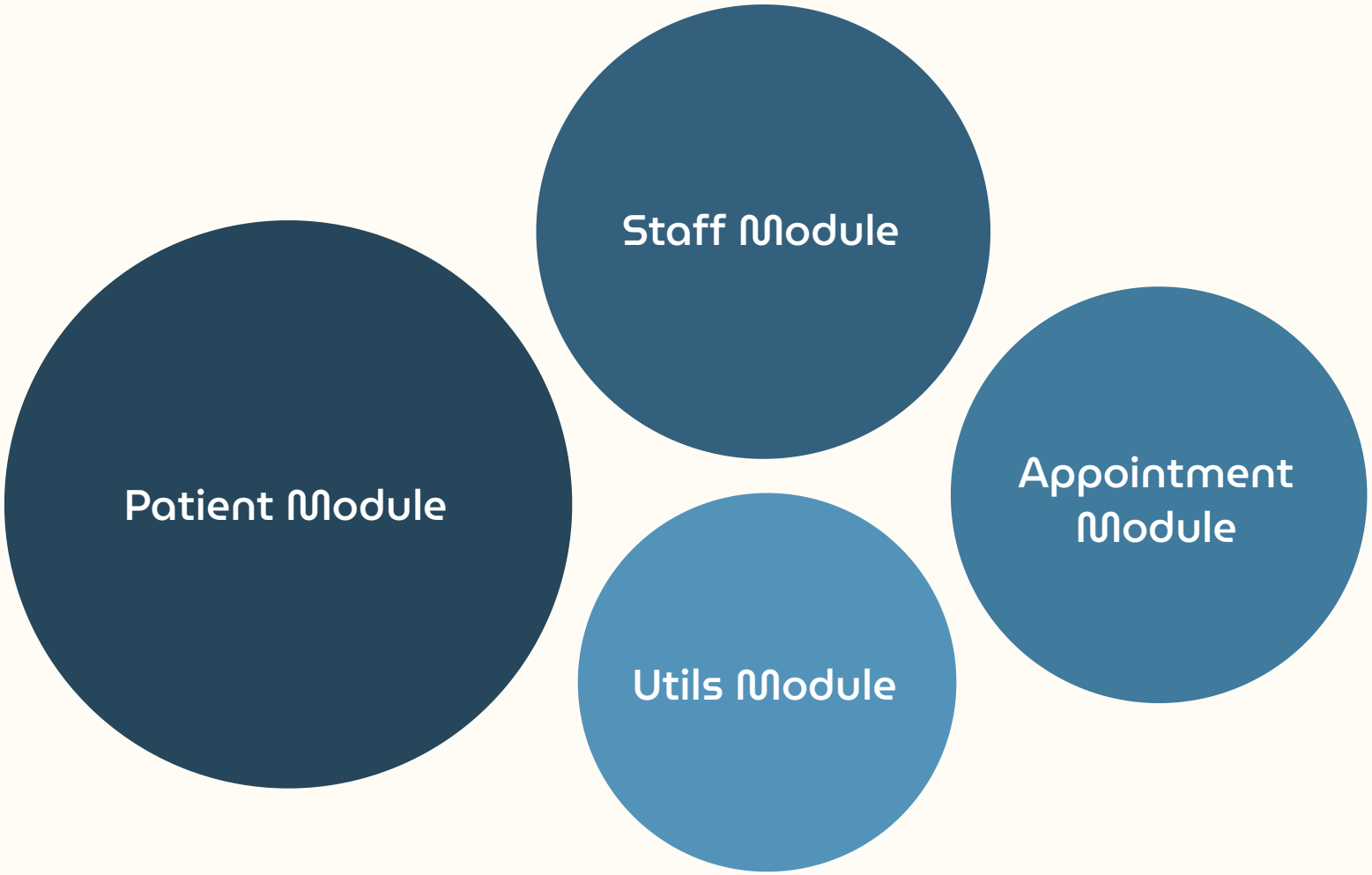
Implementation & Experimental Setup



The system is organized into four modular C source files, each responsible for a distinct domain of hospital management.

Features Implemented

- **Patient Management:** Add, update, delete, and view patient records
- **Staff Management:** Full CRUD operations for hospital staff
- **Appointment Scheduling:** Book, view, and cancel appointments

Experimental Setup

- Tested on **Windows 10** with MinGW GCC
- Data files created and updated **dynamically**
- User-friendly prompts with **error handling**

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Results & Conclusion

4

Modules

Patient, Staff, Appointment, Utils

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CRUD Domains

Fully persistent record management

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Data Loss

CSV storage ensures reliability

The system successfully manages hospital data with **minimal errors**. CRUD operations work reliably with persistent CSV storage, and the console interface is intuitive for hospital staff.

> ⓘ **Future Work:** GUI integration, multi-user support, and a database backend (e.g., SQLite) for scalability.

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References & Thank You

📚 References

GitHub: priyanshu0508/Hospital-Management-System (2025)

GeeksforGeeks: Hospital Management System in C (2023)

Scribd: Hospital Patient Record Management System (2024)

Thank You!

We appreciate your time and attention. This project demonstrates how foundational C programming can solve real-world healthcare challenges.

Questions? We welcome your feedback and suggestions.

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